Practice Problems
Calculus Physics II

Name: Key Date: _____

Chapter 60 – Imaging with Lenses

1) Which of the following terms describe lenses that are thinner at the center than at the edges? (There could be more than one correct choice.)

A) converging lenses
(B) diverging lenses
(C) concave lenses
D) convex lenses



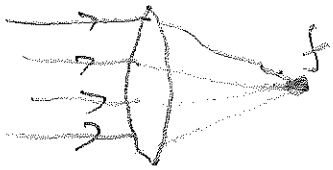
2) Which of the following terms describe lenses that are thicker at the center than at the edges? (There could be more than one correct choice.)

(A) converging lenses
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(D) convex lenses



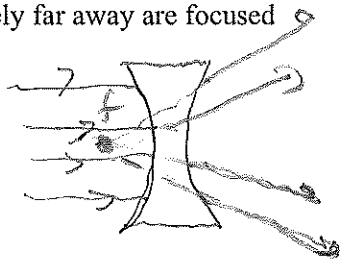
3) A convergent lens has focal length *f*. If an object is located at "infinity" (very far away), the image formed is located at a distance from the lens

A) at 2*f*.
B) between *f* and 2*f*.
(C) at *f*.
D) between the lens and *f*.
E) at infinity.



4) In a divergent lens, objects infinitely far away are focused

A) beyond the focal point.
B) at the focal point.
C) in front of the focal point.
(D) at the virtual focal point.



5) An object is placed in front of a lens if the image formed by the lens is <u>real</u>, where is the imaged formed?

(A) Opposite side of the lens as the object.
B) Same side of the lens as the object.
C) Either opposite or same side as the object.
D) Can not be determined unless the type of lens is known.

6) An object is placed in front of a lens, if the image formed by the lens is virtual, where is the imaged formed?

A) Opposite side of the lens as the object.
B) Same side of the lens as the object.
C) Either opposite or same side as the object.
D) Can not be determined unless the type of lens is known.

7) If an object is placed between a convergent lens and its focal point, the image formed is

A) virtual and upright.
B) virtual and inverted.
C) real and upright.
D) real and inverted.

8) The classroom projector forms what type of image?

A) Virtual.
B) Real.
C) Either virtual or real.

9) Starting from very far away, an object is moved closer and closer to a converging lens, stopping just before the focal length. What happens to its image formed by that lens?

A) The image gets closer and closer to the lens
B) The image gets farther and farther from the lens.
C) The image eventually changes from virtual to real.
D) The image eventually changes from real to virtual.

10) An object is placed at the focal length of a converging lens, what are the properties of the image formed

A) It is real and inverted.
B) It is virtual and upright.
C) There is no image formed the ray from the lens are parallel
D) The image is formed at the second focal point.

11) An object is placed on the left side of a thin lens, and the image of this object is formed on the left side of the lens. Which of the following statements *must* be true? (There could be more than one correct choice.)

A) The lens must be a diverging lens, and the image is upright.
B) The lens must be a converging lens, and the image is upright.
C) The image is upright and virtual.
D) The lens could be either a converging or a diverging lens, and the image is upright.
E) The image is inverted and real.

12) If the calculated magnification is negative what does this say about the image in a single lens system?

A) The image is upright
B) The image is inverted
C) The image must be larger than the object.
D) The image must be smaller than the object.

13) If the image distance is negative what does this say about the image in a single lens system?

A) The image is real.
B) The image is virtual.
C) The image is larger than the object.
D) The image is smaller than the object.

14) If the image distance is positive what does this say about the image in a single lens system?

A) The image is real.
B) The image is virtual.
C) The image is larger than the object.
D) The image is smaller than the object.

15) A leaf 3 cm long is placed 10 cm in front of a lens, the image of the leaf forms 30 cm from the lens, what is the magnification of the leaf?

$m = \frac{h_i}{h_o} = -\frac{d_i}{d_o} = -\frac{30cm}{10cm} = -3$

16) The image of a book is 40 cm long, the magnification of the lens is m = 0.6, what is the actual length of the book?

$m = \frac{h_i}{h_o}$ | $h_o = \frac{h_i}{m} = \frac{40cm}{0.6} = 67cm$

17) A box that 50 cm tall is placed in front of a lens, the image of the box is formed on the same side of the lens as the box and is 300 cm tall, what is the magnification of the box?

$m = \frac{-d_i}{d_o} = \frac{-(-300cm)}{50cm} = +6$

$d_i = -300cm$ ← Virtual

18) Given the object and image distances below, what is the magnification of the lens, is the image upright or inverted and is the image real or virtual? do = 20cm di = -80 cm

$m = \frac{-d_i}{d_o} = \frac{-(-80cm)}{20cm} = +4$

Virtual

upright

19) A 4.0 cm tall object is placed 60 cm away from a converging lens of focal length 30 cm. Sketch the ray diagram and determine the properties of the image (d_i, m, upright or inverted and real or virtual)?

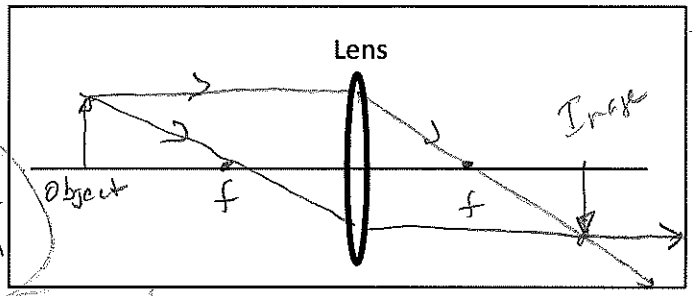
$h_o = 4cm$

$d_o = 60cm$

$f = +30cm$

$\frac{1}{f} = \frac{1}{d_i} + \frac{1}{d_o}$

$d_i = \left(\frac{1}{f} - \frac{1}{d_o}\right) = +60cm$ ← real



$m = -\frac{d_i}{d_o} = -\frac{60cm}{60cm} = -1$ ← Inverted

20) A cell phone is placed 35 cm in front of a converging lens with a focal length of magnitude 40 cm. How far is the image of this phone from the lens? Sketch the ray diagram and determine the properties of the image (d_i, m, upright or inverted and real or virtual)

$d_o = 35\text{cm}$

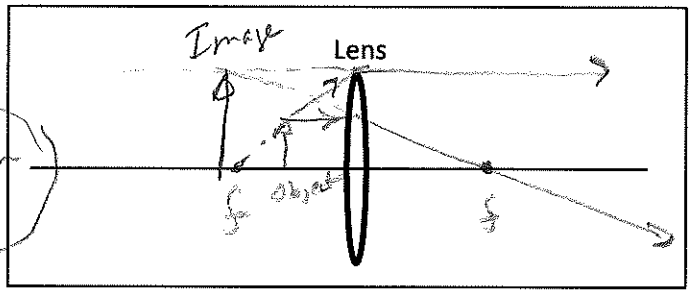
$f = +40\text{cm}$

$\frac{1}{f} = \frac{1}{d_i} + \frac{1}{d_o}$

$d_i = \frac{1}{\left(\frac{1}{f} - \frac{1}{d_o}\right)} = -280\text{cm}$ ← virtual

$m = -\frac{d_i}{d_o}$

$m = +8$ ← upright



21) A coin is 12 cm in front of a converging lens with focal length of magnitude 11 cm. Sketch the ray diagram and determine the properties of the image (d_i, m, upright or inverted and real or virtual)

$d_o = 12\text{cm}$

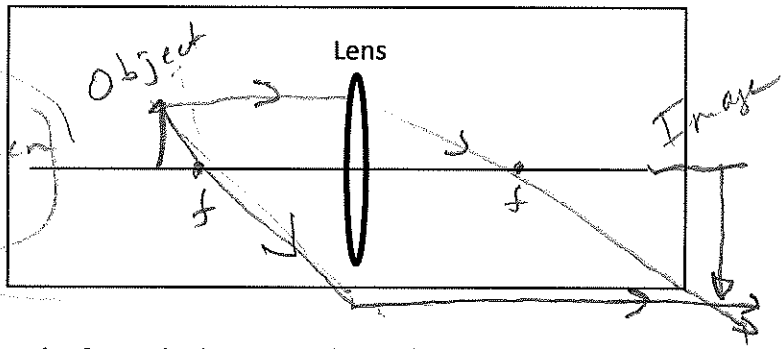
$f = +11\text{cm}$

$\frac{1}{f} = \frac{1}{d_i} + \frac{1}{d_o}$

$d_i = \frac{1}{\left(\frac{1}{f} - \frac{1}{d_o}\right)} = +132\text{cm}$ ← real

$m = -\frac{d_i}{d_o}$

$m = -11$ ← Inverted



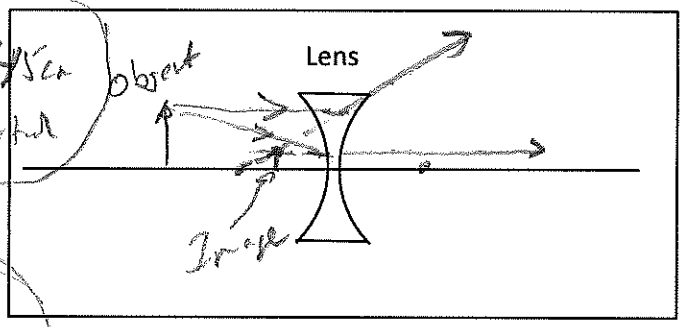
22) A coin is 12 cm in front of a diverging lens with focal length of magnitude 10 cm. Sketch the ray diagram and determine the properties of the image (d_i, m, upright or inverted and real or virtual)

$d_o = 12\text{cm}$

$f = -10\text{cm}$ ← diverging lens

$d_i = \frac{1}{\left(\frac{1}{f} - \frac{1}{d_o}\right)} = -5.45\text{cm}$ ← virtual

$m = -\frac{d_i}{d_o} = +0.45$ ← upright

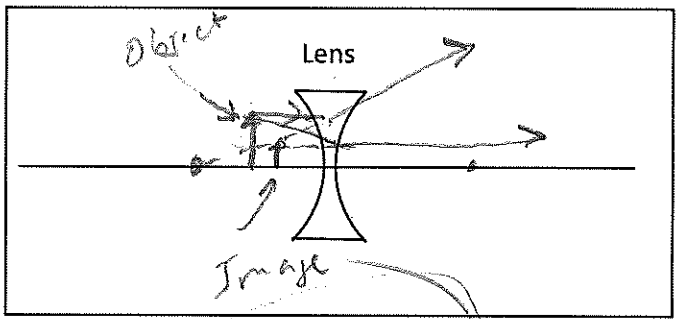


23) A page is 10 cm in front of a diverging lens with focal length of magnitude 30 cm. Sketch the ray diagram and determine the properties of the image (d_i, m, upright or inverted and real or virtual)

$f = -30\text{cm}$ ← diverging lens

$d_o = 10\text{cm}$

$d_i = \frac{1}{\left(\frac{1}{f} - \frac{1}{d_o}\right)} = -7.5\text{cm}$ ← virtual



$m = -\frac{d_i}{d_o} = +0.75$ ← upright

24) An image is 4.0 mm in front of a converging lens forms an image 2.2 mm in front of the lens, what is the focal length of the lens?

virtual image

$d_o = 4\text{ mm}$

$d_i = -2.2\text{ mm}$ (virtual)

$\frac{1}{f} = \frac{1}{d_i} + \frac{1}{d_o}$

$f = \frac{1}{\left(\frac{1}{d_i} + \frac{1}{d_o}\right)} = -4.9\text{ mm}$ ← diverging lens

25) A plant that is 4.1 cm tall is 10.3 cm from a converging lens. You observe that the image of this plant is virtual and 6.2 cm tall. What is the focal length of the lens?

$h_o = 4.1\text{ cm}$

$d_o = 10.3\text{ cm}$

$h_i = 6.2\text{ cm}$

$m = \frac{h_i}{h_o} = 1.51$

$m = -\frac{d_i}{d_o}$ ← virtual image

$d_i = -m d_o = -15.6\text{ cm}$

$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$

$f = \frac{1}{\left(\frac{1}{d_o} + \frac{1}{d_i}\right)} = +30.3\text{ cm}$ (converging lens)

26) A lab specimen is 15 mm from a converging lens. The image is 2 mm tall and 900 mm from the lens. How tall is the specimen?

Opposite side of the lens - real image

$d_o = 15\text{ mm}$

$f = 2\text{ mm}$

$d_i = +900\text{ mm}$ (real)

$m = -\frac{d_i}{d_o} = \frac{h_i}{h_o}$

$h_o = \left(\frac{d_o}{d_i}\right) h_i = 0.03\text{ mm}$

27) A wrench with a length L = 20 cm is placed at 30 cm in front of a diverging lens with a focal length of magnitude 10 cm. What is the length of the image of the wrench?

$h_o = 20\text{ cm}$

$d_o = 30\text{ cm}$

$f = -10\text{ cm}$ (diverging lens)

$d_i = \frac{1}{\left(\frac{1}{f} - \frac{1}{d_o}\right)} = -7.5\text{ cm}$

$m = -\frac{d_i}{d_o} = 0.25$

$m = \frac{h_i}{h_o}$

$h_i = m h_o = 5\text{ cm}$

28) A photograph is 10 cm tall, and the image formed by a lens produces a real image of the photograph that is 100 cm tall on a screen that is 2 m from the lens, what is the focal length of the lens required to produce this image?

$h_o = 10\text{ cm}$

$h_i = 100\text{ cm}$

$d_i = +2{,}000\text{ cm}$ (real)

$m = \frac{h_i}{h_o} = -10$ ← inverted

$m = -\frac{d_i}{d_o}$

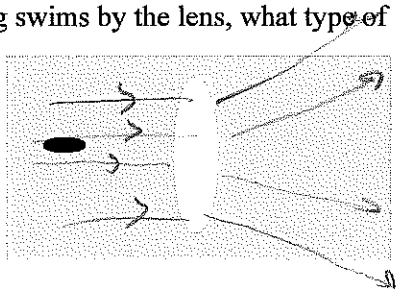
$d_o = \frac{d_i}{m} = +200\text{ cm}$

$\frac{1}{f} = \frac{1}{d_i} + \frac{1}{d_o}$

$f = \frac{1}{\left(\frac{1}{d_i} + \frac{1}{d_o}\right)} = +182\text{ cm}$

29) A hollow lens in the shape of a convergent lens is placed under water, a bug swims by the lens, what type of image of the bug will the lens create?

acts as a divergent lens



(A) Only a virtual image.
B) Only a real image.
C) It could be either real or virtual depending on where the bug is relative to the focal length of the lens.
D) There will be no image formed by the lens.

30) A glass divergent lens is placed in a beaker of oil with has the same index of refraction as the lens, what type of image could this lens produce?

A) Only a virtual image.
B) Only a real image.
C) It could be either real or virtual depending on where the bug is relative to the focal length of the lens.
(D) There will be no image formed by the lens.

no refraction of light

31) What is the power of a convergent lens that has a focal length of 40 cm?

$f = +40\,cm = +0.4\,m$

$P = \frac{1}{f} = +2.5$ Diopters

32) What is the power of a divergent lens that has a focal length of 40 cm?

$f = -40\,cm = -0.4\,m$

$P = \frac{1}{f} = -2.5$ Diopters

33) Mike's eyeglasses have a prescription of P = - 0.33 Diopters, what is the type and focal length of Mike's lenses?

$P = \frac{1}{f}$ $\quad f = \frac{1}{P} = -3\,m$

diverging lens

34) Chromatic Aberrations in lenses are due to

A) the curvature of the lens.
(B) the wavelength of light having different index of refraction.
C) the different wavelengths of light coming in at different angles.
D) the light not being white light.

35) Spherical Aberrations in lenses are due to

(A) the curvature of the lens.
B) the wavelength of light having different index of refraction.
C) the different wavelengths of light coming in at different angles.
D) the light not being white light.

36) If a leaf is placed 1.3 cm in front of a convergent lens ($f_1 = +1.25$ cm) and a second convergent lens ($f_2 = 5$ cm) is placed L = 37 cm behind the first convergent lens, what are the properties of the image formed by the lens combination?

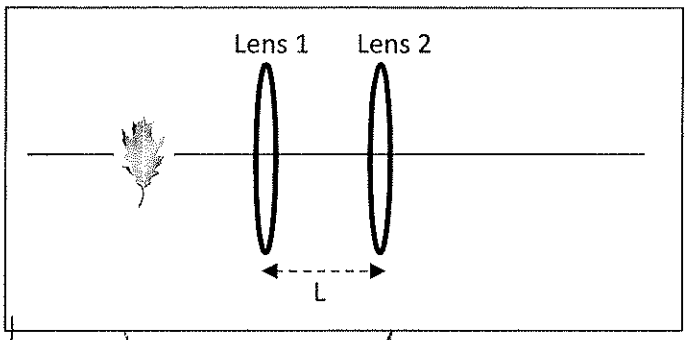


Image distance For 1st Lens

$d_{o_1} = 1.3$cm $\quad \frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$

$f_1 = 1.25$cm

$d_{i_1} = \frac{1}{\left(\frac{1}{f_1} - \frac{1}{d_{o1}}\right)} = 32.5$cm

Object For Lens is image of Lens 1

$d_{o_2} = L - d_{i_1} = +4.5$cm

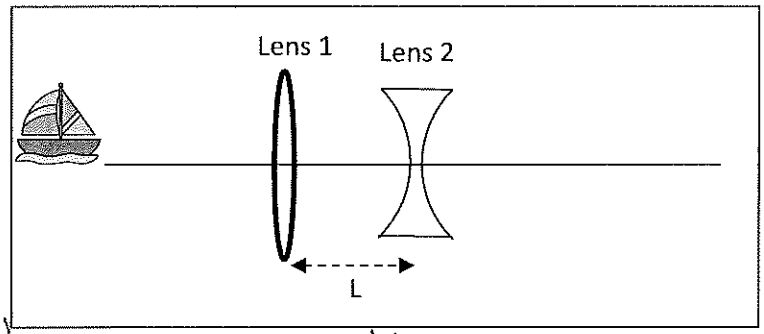
$d_{i_2} = \frac{1}{\left(\frac{1}{f_2} - \frac{1}{d_{o_2}}\right)} = -4.5$cm ↑ Virtual Image

$m_1 = \frac{d_{i_1}}{d_{o_1}} = -25$

$m_2 = -\frac{d_{i_2}}{d_{o_2}} = +10$

$m_T = m_1 m_2 = -250$ ↑ Inverted Image

37) A sailboat is 1,000 m (1×10^5 cm) from the front of a convergent lens ($f_1 = 40$ cm) and a distance L = 46 cm is a divergent lens ($f_2 = -5$ cm), what are the properties of the image formed by the lens combination?



1st lens

$d_o = 1{,}000$ m $= 1 \times 10^5$cm

$f_1 = 40$cm

$d_{i_1} = \frac{1}{\left(\frac{1}{f_1} - \frac{1}{d_{o1}}\right)} = 40$cm

$m_1 = -\frac{d_{i_1}}{d_{o_1}} = -4 \times 10^{-4}$

2nd lens

$d_{o_2} = L - d_{i_1} = 6$cm

$f_2 = -5$cm

$d_{i_2} = \frac{1}{\left(\frac{1}{f_2} - \frac{1}{d_{o_2}}\right)} = -2.73$ ↑ Virtual Image

$m_2 = -\frac{d_{i_2}}{d_{o_2}} = +0.45$

$m = m_1 m_2 = -1.82 \times 10^{-4}$ ↑ Inverted

KEY

1) B and C
2) A and D
3) C
4) D
5) A
6) B
7) A
8) B
9) B
10) C
11) C and D
12) B
13) B
14) A
15) -3
16) 67 cm
17) + 6
18) +4
19) di = + 60 cm real m = -1 inverted
20) di = - 280 cm virtual m = + 8 upright
21) di = + 132 real m = - 11 inverted
22) di = - 5.45 cm virtual m = + 0.45 upright
23) di = - 7.5 cm virtual m = + 0.75 upright
24) f = - 4.9 mm divergent lens
25) f = + 30.3 cm convergent lens
26) ho = 0.03 mm
27) hi = 5 cm
28) f = + 0.182 m convergent lens
29) A
30) D
31) + 2.5 D
32) – 2.5 D
33) – 3 m
34) B
35) A
36) d_{i1} = + 32.5 cm and d_{i2} = - 45 cm
37) d_{i1} = + 40 cm and d_{i2} = - 2.73 cm